



North South University

Department of Electrical & Computer Engineering

Individual project

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North South University

Department of Electrical & Computer Engineering

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Smart Home Automation System with Integrated Delivery and Security Monitoring Using CORTEX-M

Abstract

Home automation systems are becoming essential in modern homes due to their ability to enhance convenience, security, and energy efficiency. Leveraging ARM Cortex-M microcontrollers, known for their low power consumption and high processing capabilities, is an optimal solution for managing multiple devices in a home. This research focuses on developing an assembly program that uses an 8-bit control register, `CONTROL\_REG`, where each bit controls a specific home device (lights, heating, security systems, etc.). The project aims to control devices such as lights, heating, and security systems using bitwise operations like AND, ORR, BIC, and EOR to perform tasks such as turning devices on or off, toggling states, and checking device statuses. Specifically, the program will turn on the living room lights and heating system, turn off the air conditioning, toggle the security alarm, and check the state of the garage door to close it if open. The expected contribution includes demonstrating the efficiency of Cortex-M microcontrollers in home automation through practical and theoretical exploration of bitwise manipulation, offering a streamlined and energy-efficient control system that can be applied to other embedded systems.

Keywords: - ARM Cortex-M, microcontroller, home automation, bitwise operations, CONTROL_REG, embedded systems, assembly programming, device control, energy efficiency.

Contents

Abstract	i
Table of Contents	ii
List of Figures	iv
List of Tables	v
1 Introduction	1
1.1 LoremIpsum ... 1.2	1
Motivation 1.3	2
ResearchQuestions 1.4	2
Outline	2
2 BackgroundandRelatedWork	3
2.1 Task2	3
Datasets 2.3	3
Models.	4
3 LoremIpsumDolorSitAmet	4
3. IntroductionLorem	4
1 Ipsum Dolor Sit Amet	4
3. Experiments	4
2 Conclusions	
3.	
4 Dataset	5

3.1 Lorem Ipsum Dolor Sit Amet	5
4	6
5 Measurement	6
5.1 Introduction.	6
5.2 Metric A	

6	ProposedWorkandTimeline	7
6.1	Lorem Ipsum Dolor Sit Amet	7
6.1.1	Description	7
6.1.2	TimeEstimate	7
7	ExpectedContributions	8

List of Figures

1.	Lorem ipsum dolor sit amet, consectetur adipiscing elit.	1
1	Lorem ipsum dolor sit amet, consectetur adipiscing elit.	3
2.		
1		

List of Tables

Chapter 1

Introduction

1.1 Lorem Ipsum

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

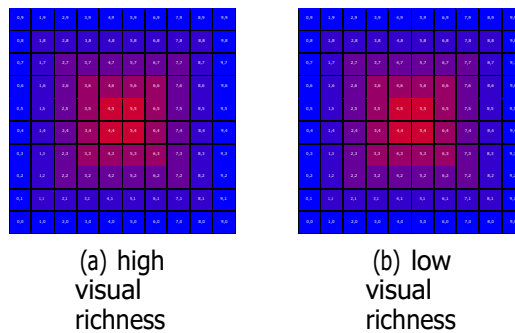


Figure 1.1: Lorem ipsum dolor sit amet, consectetur adipiscing elit.

1.2 Motivation

1.3 ResearchQuestions

1. Lorem ipsum dolor sit amet, consectetur adipiscing elit?
2. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat?
3. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur?

1.4 Outline

Chapter 2

Background and Related Work

2.1 Tasks

2.2 Datasets

2.3 Models

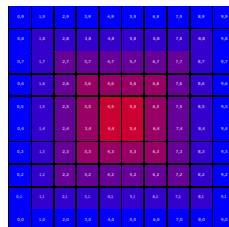


Figure 2.1: Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Chapter 3

Lorem Ipsum Dolor Sit Amet

3.1 Introduction

3.2 Lorem Ipsum Dolor Sit Amet

3.3 Experiments

3.4 Conclusions

Chapter 4

Dataset

4.1 Lorem Ipsum Dolor Sit Amet

Chapter 5

Measurement

5.1 Introduction

5.2 MetricA

Chapter 6

Proposed Work and Timeline

6.1 Lorem Ipsum Dolor Sit Amet

6.1.1 Description

6.1.2 TimeEstimate

Chapter 7

Expected Contributions

Bibliography